

Lagos, Nigeria
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Ayomide Wilfred Adeyemi

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[Portfolio](#)

Backend engineer with 4+ years shipping production NestJS / Node.js platforms at Nigerian startups — REST APIs, real-time messaging, payments, and logistics. Currently architecting a logistics + B2B integration platform serving 1000+ concurrent users, with cross-platform mobile contributions across React Native and Flutter clients. Comfortable owning systems end-to-end and equally comfortable supporting an existing roadmap.

Work Experience

Software Engineer

Backend team + Mobile team

Independent Purchasing Company

Lagos, Nigeria

Sep 2025 – Present (Full-Time)

[Relay Pilot](#) is a logistics platform connecting businesses, riders, and B2B integration partners across Nigeria.

- Architected a modular backend system with 100+ domain modules using NestJS, implementing clean architecture principles across logistics, payments, authentication, orders, and notifications
- Built multi-provider payment processing system integrating Paystack and 9JaPay with webhook signature verification, virtual account management, idempotency controls, and MongoDB ACID-compliant transactions — processing millions in financial transactions at 99.99% data consistency
- Scaled real-time logistics tracking using Redis-backed Socket.IO adapter with Redis geospatial indexing, enabling multi-instance WebSocket deployments with 100MB payload optimization for live rider location updates across 1000+ concurrent connections.
- Built an enterprise real-time messaging REST API + WebSocket platform with Socket.IO — multi-device support, distributed conversation management, push/SMS/email notifications, and S3 presigned-URL attachment uploads, serving 10,000+ concurrent users with sub-second latency.
- Shipped the B2B Integration REST API — a self-serve platform letting external partners (food aggregators, marketplaces, e-commerce stores) push delivery orders into Relay Pilot. Sandbox + live API keys, quote-based dispatch, HMAC-signed webhook delivery, and pre-paid wallet billing.
- Set up a distributed BullMQ + Redis queue system across 10+ specialised job queues for settlements, notifications, and fan-out — including a manager-to-rider broadcast feature using keyset cursor pagination + bulk receipt insert to deliver one announcement to thousands of active riders per logistics company under sub-second latency.
- Contributed to the platform's React Native (Expo) and Flutter mobile clients — real-time chat with Socket.IO reconnect handling, paginated contact picker with debounced search, BullMQ-fed in-app notifications, and the EAS Android build pipeline.

Software Engineer

Backend team

Luftsocial

Upwork

May 2025 – Sep 2025 (Freelance)

Social-media management platform for creators and agencies — scheduling, analytics, and cross-platform engagement.

- Led the real-time messaging module (NestJS + Socket.IO) — designed 20+ typed event handlers, room-based broadcasting, multi-device support (up to 8 concurrent connections per user), offline message sync, and full lifecycle features (typing indicators, read receipts, emoji reactions). Sole owner of the module across the project's lifetime
- Owned the subscription-plans domain — shipped free-plan auto-assignment for new signups, auto-fallback to free on expiry, currency-aware billing, and Paystack subscription integration with redirect handling
- Built the invoice email queue + Bull Board admin UI on BullMQ — gave operations visibility into queue depth, failures, and retry attempts across the project's job queues
- Shipped a task-based chat rooms feature with multi-assignee support — wiring the messaging service into the task-management domain so team members can collaborate per-task in dedicated rooms
- Contributed to the notifications module alongside the lead engineer — multi-channel delivery (email, push, in-app) plumbed into NestJS event handlers
- Contributed to the team's containerised deployment pipeline — Docker multi-environment config (dev / staging / prod) with health checks
- Integrated LangChain and OpenAI with custom credit tracking, subscription-based usage limits, and real-time consumption monitoring — maintaining 99.8% service availability across AI-powered features

Software Engineer

Backend team

Verify Your Customer

Lagos, Nigeria

Feb 2025 – Present (Contract)

KYC platform that detects forged IDs and verifies identity documents.

- Engineered a webhook-delivery system with custom CA-certificate pinning + public-key verification on outbound HTTPS — eliminating MITM attack vectors on the most security-sensitive surface in a KYC product. Per-attempt verification logging + exponential-backoff retry against transient failures
- Built an event-driven webhook pipeline on Bull Queue + Redis with a Bull Board admin UI for ops visibility, processing 1,000+ asynchronous jobs at 99.9% delivery reliability
- Implemented authentication covering JWT, role-based access, and Google OAuth for the platform's customer-facing dashboard + machine-to-machine REST API
- Designed a Sequelize-based DAO abstraction layer with dynamic query building, pagination, and transaction support — reducing query code duplication ~60% across 12+ models
- Integrated Stripe Payment Intents API with metadata tracking and automated subscription activation; processes monthly billing for active customers

Software Engineer

Backend team

Uwana Energy

Lagos, Nigeria

June 2024 – January 2025*Solar-finance lending platform connecting providers, lenders, and end customers.*

- Architected a Node.js REST API using clean layered architecture (Controllers, Services, DAO) organising 50+ controllers across role-scoped domains (customer, provider, admin, lender) — enabling parallel development across the team
- Built a Sequelize ORM data layer with a BaseDAO abstraction providing dynamic query building, pagination, and transaction support — reducing database code duplication ~70% across 40+ models. Dialect-agnostic schema for portability between local SQLite and production MySQL
- Integrated Pystack payment gateway with webhook signature verification, dynamic fee calculation, and escrow management — processing customer-to-provider settlements at 99.9% uptime
- Designed a transactional email service with Nodemailer + Mustache templates supporting 15+ email types (escrow deposits, KYC, payment confirmations, subscription events)
- Shipped an end-to-end GitLab CI/CD pipeline with Docker containerisation, automated deployments to AWS ECS, and image cleanup — reducing deploy time ~75% across dev / staging / prod

Software Engineer

Backend team

Sokozuri Africa

Lagos, Nigeria

August 2024 – May 2025 (Contract)*Multi-vendor e-commerce marketplace.*

- Designed and managed 36+ Sequelize migrations across 15+ interconnected entities supporting multi-vendor operations (vendors, customers, admin, products, orders, deliveries, reviews, categories)
- Built a real-time Socket.IO messaging system spanning three role channels (customer, vendor, admin) with user presence, online-status tracking, and per-conversation rooms — enabling instant communication across the marketplace
- Engineered a Redis caching layer with automatic response interception and a 1-hour TTL — reducing database load ~60% and improving median API response from ~500 ms to ~150 ms
- Integrated Pystack payments with webhook verification + AWS S3 / DigitalOcean Spaces file storage using pre-signed URLs (30 MB uploads, 10,000+ product images at scale)
- Built a transactional email + OTP system with template engine, sending 5,000+ emails/day for user verification and order notifications at ~98% deliverability

Backend engineering for Pisonitsha — a student-management platform (admissions, courses, payments, notifications).

- Built a modular Node.js / Express backend in MVC layout (auth, courses, students, teachers, notifications) with Sequelize + PostgreSQL — schema design, model relationships, and migration workflow
- Implemented automatic matriculation-number generation — class-based sequential numbering tied to enrolment data so new students get a deterministic ID on creation
- Integrated Paystack for online tuition/fee payments — transaction initialisation, verification, and error handling against the Paystack REST API
- Added Swagger / OpenAPI documentation for the public endpoints, with the interactive UI served on the dev environment for the frontend team
- Containerised the app with Docker + docker-compose for database services and environment-based configuration; implemented bcrypt password hashing, JWT auth, and CORS protection
- Mentored junior engineers and interns on backend setup, database design, and Git workflows

Technologies and Languages

- Languages: JavaScript, TypeScript, Node.js, Python.
- Storage: MySQL, Postgres, MongoDB, S3, Redis, Minio.
- Orms: Sequelize, TypeORM, Prisma, Mongoose, sqlalchemy.
- Platforms: Azure, Docker, Amazon Web Services (AWS), Digital Ocean, Vercel, Heroku, Notion.
- Mobile: React Native (Expo), Flutter
- Frameworks: Express, Nest.js, FastAPI, Multer, Socket.IO.
- Payments: Paystack & Stripe API integration.
- Tools: Postman, Git.

Education and Certifications

- **B.Eng. Electrical and Electronics Engineering**, Kwara State University, Nigeria. **2018 – 2022**
- **Quantum Software Engineering - Quantum** **2024**
- **AWS APAC Solutions Architecture** **2024**
- **Azure Kubernetes Service - KodeKloud** **2024**
- **Cisco Software Engineering - Cisco** **2022**
- **Accenture North America Virtual Coding Experience - Forage** **2022**

Projects and Blogs

- Qoinpal Crypto — A Go-based disposable cryptocurrency wallet generator for Bitcoin and Ethereum networks. Implements ECDSA keypair generation, multi-stage address hashing (SHA256, RIPEMD160, Keccak256), BadgerDB local persistence, and a REST API via the Gin framework. See the source on [Github](#).
- [How to build a QR code generator for Urls with Node.js, Next.js, and azure blob storage](#)
- [How to run a postgres database in azure kubernetes service and integrate it with a Node.js express application](#)
- [A guide to connecting to postgresql with supabase and sequelize orm.](#)

Interests

- **Artificial Intelligence and Machine Learning** – Passionate about building intelligent systems and exploring AI's transformative potential across industries.
- **Microservices Architecture** – Advocating for modular, maintainable system designs for seamless deployment and scaling.
- **DevOps and CI/CD Pipelines** – Committed to automating workflows and ensuring seamless development-to-production transitions.
- **Software Design Patterns** – Leveraging proven design principles for building robust, maintainable software solutions.
- **Serverless Architectures** – Exploring the potential of serverless technologies for cost-effective, event-driven solutions.
- **Exploring Blockchain Technology** – Staying updated on the latest advancements and innovations in decentralized finance and blockchain ecosystems.
- **IoT and Smart Devices** – Enthusiastically exploring the intersection of software and hardware in building connected systems for smarter living.